

Convex Optimization for Contraction-Based Control: An Energy-Based Effective Space Approach ★

Myeongseok Ryu^{*} Kyunghwan Choi^{*} Sesun You^{**}

^{*} Korea Advanced Institute of Science and Technology (KAIST),
Daejeon 34051, South Korea (e-mail: {dding_98,kh.choi}@kaist.ac.kr)

^{**} Keimyung University, Daegu 42601, South Korea (e-mail:
yousesun@kmu.ac.kr)

Abstract: In this paper, we propose a computationally efficient and numerically robust contraction metric calculation method for contraction-based control design. Conventional Linear Matrix Inequality (LMI) frameworks for contraction metric calculation suffer from high computational complexity and scale-invariance issues of the objective function which minimizes the condition number of the contraction metric. To address these issues, the contraction metrics are projected onto the trajectory error vector space leading to a simplified convex optimization framework within a lower-dimensional effective space. Furthermore, the energy-based constraints are incorporated preventing the contraction metrics from collapsing or exploding, which ensures minimum required control effort. In addition, convex input saturation constraints are considered to handle practical limitations of the actuators. The effectiveness and the feasibility of the proposed method is validated through numerical simulations via Lorenz system.

Keywords: Contraction theory, convex optimization, input saturation, energy-based control

NOTATION

In this study, the following notation is used: Vector and matrix entries are denoted as $\mathbf{x} = [x_i]_{i \in \{1, \dots, n\}}$ and $\mathbf{A} = [a_{ij}]_{i \in \{1, \dots, n\}, j \in \{1, \dots, m\}}$, respectively. The symmetric part of a matrix is denoted as $\text{sym}(\mathbf{A}) := \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ Tsukamoto et al. (2021a). The identity matrix of size n is denoted by $\mathbf{I}_n$, and the zero matrix of size $n \times m$ is denoted by $\mathbf{0}_{n \times m}$. For $\mathbf{A} \in \mathbb{R}^{n \times m}$, the vectorization of $\mathbf{A}$ is denoted as $\text{vec}(\mathbf{A}) := (\text{row}_1(\mathbf{A}^\top), \dots, \text{row}_m(\mathbf{A}^\top))^\top \in \mathbb{R}^{nm}$.

1. INTRODUCTION

1.1 Motivation

Contraction Theory

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Conventional Approaches

- LMI framework for contraction metric calculation

Limitations

- Computational complexity of LMI framework for contraction metric calculation
- No consideration of input saturation issue

To the best of our knowledge, there has been no previous work that projects the contraction constraints onto the

error space to reduce the computational complexity of the contraction metric calculation. Moreover, input saturation has not been considered in the conventional contraction theory based control design. Hence, the main objective of this paper is to simplify the contraction metric calculation and to incorporate the input saturation constraint into a convex optimization framework for contraction metric calculation.

1.2 Contributions

The main contributions of this paper are

- The contraction metrics can be calculated in a lower-dimensional effective space by projecting the contraction constraints onto the trajectory error vector space, which significantly reduces the computational complexity of the LMI framework for contraction metric calculation;
- Scale-invariance issues of the objective function in the conventional LMI framework are resolved by introducing energy-based constraints that prevent the contraction metrics from collapsing or exploding, which ensures minimum required control effort;
- Saturation constraint is incorporated into the convex optimization framework, which handles practical limitations of the actuators; and
- Robust feasibility of the proposed method is enhanced by ... slack variables

2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorems. The detailed ex-

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planation of contraction theory can be found in Lohmiller and Slotine (1998); Tsukamoto et al. (2021b).

2.1 Contraction Property and Contraction Metric

Consider that we have the following deterministic nonlinear system:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ denotes the nonlinear system function. Additionally, consider differential dynamics of the system (1) (Kirk, 2004, Chap 4), i.e., $\frac{d}{dt}\delta\mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\delta\mathbf{x}$.

According to [...], the system (1) is said to be *contracting*, i.e., all trajectories converge incrementally exponentially to a unique trajectory, if the following condition holds:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\frac{\partial \mathbf{f}}{\partial \mathbf{x}}) \preceq -2\alpha\mathbf{M}, \quad (2)$$

where $\mathbf{M}$ denotes *contraction metric* and $\alpha \in \mathbb{R}_{>0}$ denotes the contraction rate. The contraction metric $\mathbf{M}$ is symmetric positive definite matrix and satisfies $\bar{m}\mathbf{I}_n \succeq \mathbf{M} \succeq \underline{m}\mathbf{I}_n$ with $\bar{m}, \underline{m} \in \mathbb{R}_{>0}$.

It is notable that the initial conditions are exponentially forgotten in the contracting system Lohmiller and Slotine (1998).

2.2 Perturbed Contracting System

If the system (1) is subjected to bounded disturbances, the system can be expressed as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{d}(t), \quad (3)$$

where $\mathbf{d}(t) \in \mathbb{R}^n$ denotes the external disturbances with $\|\mathbf{d}(t)\| \leq \bar{d}$ where $\bar{d} \in \mathbb{R}_{>0}$.

The following theorem states that all solutions of the perturbed contracting system (3) exponentially converge to a bounded ball of the solution of the nominal contracting system (1).

Theorem 1. If the system (1) is contracting with the contraction metric $\mathbf{M}$ which satisfies the condition (2), the path integral $V_i(\mathbf{q}, \delta\mathbf{q}, t) = \int_{\xi_1}^{\xi_2} \|\Theta(\mathbf{q}, t)\delta\mathbf{q}\|$, where $\Theta(\mathbf{q}, t) := \mathbf{M}^{1/2}$, and ξ_1 and ξ_2 are the solutions of (1) and (3), respectively, and $\mathbf{q}$ is the virtual state variable, exponentially converges to a bounded ball. Especially, the following inequality of a bounded error ball of the solutions of (1) and (3) holds:

$$\|\xi_1(t) - \xi_2(t)\| \leq \frac{V_i(0)}{\sqrt{\bar{m}}} \exp(-\alpha t) + \frac{\bar{d}}{\alpha} \sqrt{\chi} (1 - \exp(-\alpha t)), \quad (4)$$

where $\chi := \bar{m}/\underline{m}$ denotes the condition number of $\mathbf{M}$, yielding the steady state error bound of $\frac{\bar{d}}{\alpha}\chi$.

Proof. See the proof of Theorem 2.4 in Tsukamoto et al. (2021b).

3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat}(\mathbf{u}) + \mathbf{d}(t), \quad (5)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables, $\mathbf{u} \in \mathbb{R}^m$ denotes the control inputs, and $\mathbf{d} := \mathbf{d}(t) \in \mathbb{R}^n$ denotes

the bounded external disturbances, i.e., $\|\mathbf{d}\| \leq \bar{d}$, and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ and $\mathbf{B} := \mathbf{B}(\mathbf{x}, t)$ denote system functions. The input saturation function is denoted by $\text{sat}(\cdot)$ and can be defined as any convex function which is physically feasible.

We assume that some desired control input $\mathbf{u}_d \in \mathbb{R}^m$ is pre-designed to achieve the desired trajectory $\mathbf{x}_d \in \mathbb{R}^n$ under ideal conditions, i.e., $\mathbf{d} = 0$ and $\|\mathbf{u}_d\| \leq \bar{u}$ where $\bar{u} \in \mathbb{R}_{>0}$ denotes the maximum allowable input magnitude by simulating the following desired system:

$$\frac{d}{dt}\mathbf{x}_d = \mathbf{f}_d + \mathbf{B}_d\mathbf{u}_d, \quad (6)$$

where $\mathbf{f}_d := \mathbf{f}(\mathbf{x}_d, t)$ and $\mathbf{B}_d := \mathbf{B}(\mathbf{x}_d, t)$.

The error dynamics between the system (5) and the desired system (6) can be expressed as follows:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \mathbf{f} - \mathbf{f}_d + \mathbf{B} \text{sat}(\mathbf{u}) - \mathbf{B}_d\mathbf{u}_d + \mathbf{d}. \quad (7)$$

Using the state-dependent coefficient (SDC) formulation Dani et al. (2015), i.e., $\mathbb{A} := \int_0^1 \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right] (c\mathbf{x} + (1-c)\mathbf{x}_d) dc$, the error dynamics (7) can be linearized as follows:

$$\begin{aligned} \frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) &= \underbrace{\mathbf{f} - \mathbf{f}_d}_{=\mathbb{A}(\mathbf{x}, \mathbf{x}_d)(\mathbf{x} - \mathbf{x}_d)} + \underbrace{(\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=\mathbb{B}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}} \\ &= \mathbb{A}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d} \end{aligned} \quad (8)$$

To simplify the control design, we use the following state feedback control law:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (9)$$

where $\mathbf{R} \in \mathbb{R}^{m \times m}$ denotes a positive definite weighting matrix for the control input. Substituting the control law (9) into the error dynamics (8), we obtain:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \left(\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \right) (\mathbf{x} - \mathbf{x}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}, \quad (10)$$

where $\phi(\mathbf{u}) := \text{sat}(\mathbf{u}) - \mathbf{u}$.

Finally, the control objective is to obtain the contraction metric $\mathbf{M}$ in (9) which guarantees the system (5) to be contracting in the presence of input saturation.

4. CONVENTIONAL LMI-APPROACH

In [...], Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to obtain the contraction metric $\mathbf{M}$ by solving the following LMI optimization problem. Ignoring the input saturation effect, the contraction constraint of the error dynamics (10) can be expressed as follows:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\mathbb{A}) - 2\mathbf{M}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \preceq -2\alpha\mathbf{M}. \quad (11)$$

Using the variable transformation $\mathbf{M} =: \mathbf{W}^{-1}$, we have the following LMI optimization problem:

$$\begin{aligned} &\min_{\mathbf{W}, \xi, \nu} \xi + \lambda\nu \\ \text{s.t. } &\begin{cases} -\frac{d}{dt}\bar{\mathbf{W}} + 2\text{sym}(\bar{\mathbf{A}}\bar{\mathbf{W}}) - 2\nu\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \preceq -2\alpha\bar{\mathbf{W}}, \\ \mathbf{I}_n \preceq \bar{\mathbf{W}} \preceq \xi\mathbf{I}_n, \end{cases} \end{aligned} \quad (12)$$

where $\bar{\mathbf{W}} := \nu\mathbf{W}$ and $\nu \in \mathbb{R}_{>0}$ is a normalizing term of $\mathbf{M}$. Using $\lambda \in \mathbb{R}_{>0}$, the magnitude of the contraction metric $\mathbf{M}$ can be adjusted.

As well known, the computational complexity of solving the above LMI optimization problem is $\mathcal{O}(n^6)$ [..], which can be a critical issue for high-dimensional systems. Moreover, since CV-STEM penalizes the magnitude of the control input indirectly by adjusting $\mathbf{M}$, the input saturation constraint is not explicitly considered in the optimization problem.

[MS: Add more!]

Remark 2. [MS: Mathematical Issue]

5. MAIN RESULTS

Since in the control law (9), the contraction metric $\mathbf{M}$ is only applied to the trajectory error vector $\mathbf{e} := \mathbf{x} - \mathbf{x}_d$, we can project the contraction constraints onto the trajectory error vector space, leading to a simplified convex optimization framework within a lower-dimensional effective space. Hence, pre- and post-multiply $\mathbf{e}^\top$ and $\mathbf{e}$ to the contraction constraint (11), respectively, yielding:

$$\mathbf{y}^\top (-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \quad (13)$$

where we define $\mathbf{y} := \mathbf{M}\mathbf{e}$. Note that $\frac{d}{dt}\mathbf{M} \approx \frac{1}{T_s}(\mathbf{M} - \mathbf{M}_{\text{pre}})$ where T_s denotes the sampling time and $\mathbf{M}_{\text{pre}}$ denotes the contraction metric obtained at the previous time step.

The convex input saturation function $\text{sat}(\cdot)$ preserves the convexity on $\mathbf{y}$, since $\mathbf{u}$ is linear with respect to $\mathbf{y}$.

As aforementioned in Section 4, the scale-invariance issue of the objective function which only minimizes the condition number of $\mathbf{M}$ should be resolved. To this end, energy-based constraint is introduced to prevent the contraction metric $\mathbf{M}$ from collapsing or exploding, which ensures minimum required control effort.

$$(\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e})^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \quad (14)$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\geq 0}$ denote allowable energy variation and a slack variable to enhance the robust feasibility of the optimization problem.

Thus, the following convex optimization problem can be formulated to obtain the contraction metric $\mathbf{M}$ considering the input saturation effect:

$$\begin{aligned} \min_{\mathbf{y}} & \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} + \lambda s \\ \text{s.t.} & \begin{cases} (13), (14), \\ \text{sat}(\mathbf{y}) \leq 0, \end{cases} \end{aligned} \quad (15)$$

For better intuition, in Fig. 1, the constraints are illustrated in two-dimensional space. The minimization of the condition number of $\mathbf{M}$ is equivalent to minimizing the angle between $\mathbf{y}$ and $\mathbf{e}$ as depicted in Fig. 1. The contraction constraint (13) and the input saturation constraint $\text{sat}(\mathbf{y}) \leq 0$ restrict the feasible space of $\mathbf{y}$ in the outside of blue eclipse and the inside of red eclipse, respectively.

The green lines represent the energy-based constraint (14).

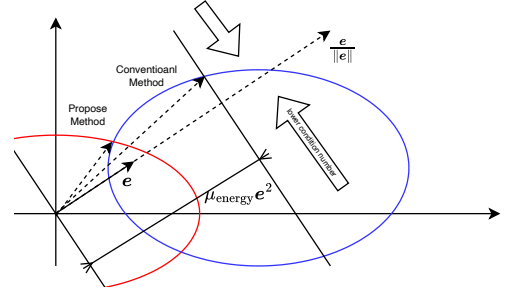


Fig. 1. Illustration of the optimization problem in two-dimensional space.

Table 1. Controllers for comparative study.

Description	
(C ₁) [—]	Proposed controller
(C ₂) [—]	CV-STEM (Tsukamoto and Chung (2021))
(C ₃) [—]	Desired controller, i.e., $\mathbf{u} = \mathbf{u}_d$

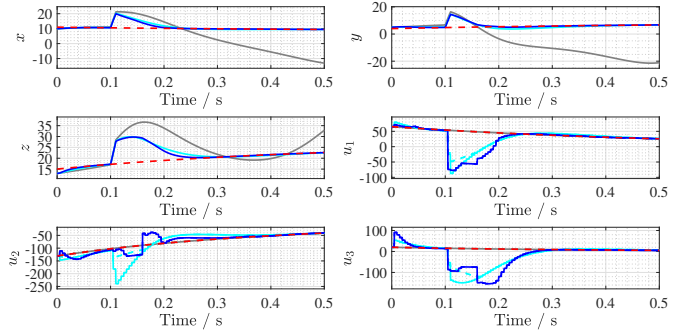


Fig. 2. Numerical validation results of (C₁) [—], (C₂) [—], (C₃) [—], and desired trajectory x_d [—].

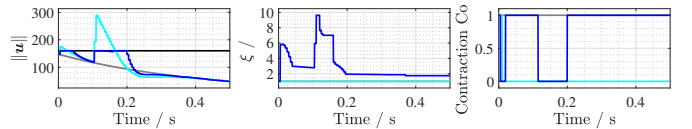


Fig. 3. Optimization results of (C₁) [—] and (C₂) [—].

6. NUMERICAL VALIDATION

6.1 Numerical Validation Set Up

We employed the Lorenz system [MS: REF] to validate the proposed method which is represented as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \xi} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}}_{=: f(\xi)} + \underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: u} + \underbrace{\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}}_{=: d}, \quad (16)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters.

6.2 Numerical Validation Results

7. CONCLUSION

In this study, ...

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Place acknowledgments here.

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